

Question:

A 10-month-old boy is brought to the emergency department by his parents because he has a high fever and severe cough. His fever started 2 days ago and his parents are concerned as he is now listless and fatigued. He had a similar presentation 5 months ago and was diagnosed with pneumonia caused by *Staphylococcus aureus*. He has been experiencing intermittent diarrhea and skin abscesses since birth. The child had an uneventful birth and the child is otherwise developmentally normal. Analysis of this patient's sputum reveals acute angle branching fungi and a throat swab reveals a white plaque with germ tube forming yeast. Which of the following is most likely to be abnormal in this patient? Answer Choices: A. Autoimmune regulator function B. Lysosomal trafficking C. NADPH oxidase activity D. Thymus development

Reasoning + Answer:

<think> The patient's symptoms of fever, cough, and listlessness, along with the presence of acute angle branching fungi and white plaque with germ tube forming yeast in the sputum and throat swab, suggest immunodeficiency or recurrent infections, which could be due to a defect in the thymus development as the thymus is crucial for the development of T-cells in the immune system, and any defect in its development can lead to recurrent bacterial and fungal infections in childhood. </think> <answer> D . Thymus development </answer>

